A COMPARISON OF INFLATION FORECASTING MODELS

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INTRODUCTION

Accurate **inflation forecasting** is vital for setting **monetary and fiscal policy** for banks and governments, influencing consumers directly through **wages, investments,** and more. However, inflation is shaped by complex, nonlinear forces: global markets, supply chains, and policies, that traditional models often fail to capture.

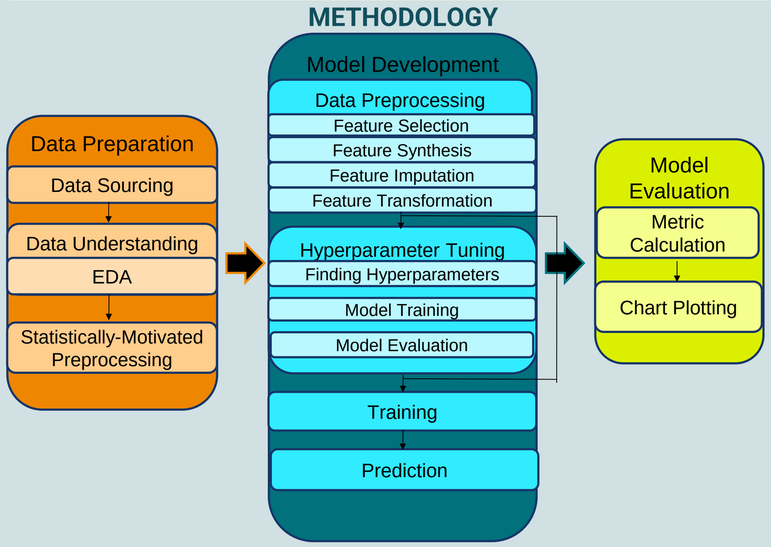
OBJECTIVE

We aim to **automate U.S. inflation forecasting**, via the **PCEPI index** [1], by replacing manual, theory-based methods with data-driven models. We implement and evaluate **over 15 models:** spanning **classical** (ARDL [2], VAR [3], SARIMAX [4]), **machine learning** (Random Forest [5], XGBoost[6]), and **deep learning** (LSTM [7], GRU [8], TFT [9], N-HiTS[10]) categories, using macroeconomic data and walk-forward evaluation across multiple forecast horizons.

METHODS AT A GLANCE

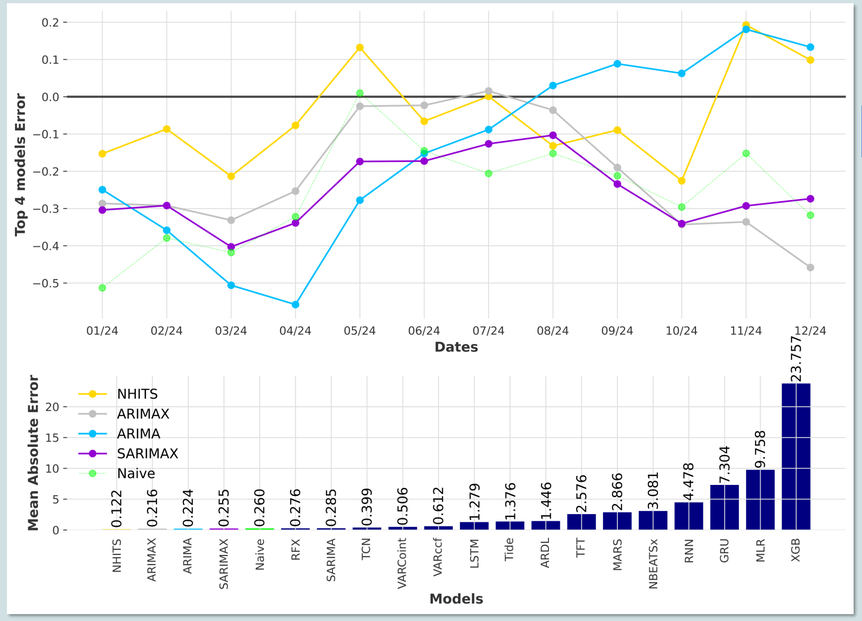
Category	Model	Key Idea
Classical	ARDL	Captures lagged effects of variables.
	VAR	Models dependencies between time series.
	ARIMA	Uses past values and errors; assumes stationarity.
	SARIMA	Adds seasonality to ARIMA.
	ARIMAX	ARIMA with external variables.
	SARIMAX	Seasonal ARIMA with exogenous inputs.
Machine Learning	Random Forest	Tree ensemble for non-linear prediction.
	XGBoost	Boosted trees for strong tabular performance.
	MARS	Regression using adaptive splines.
Deep Learning	RNN	Learns from sequence history.
	GRU	RNN variant with gated memory.
	LSTM	Captures long-term dependencies with gates.
	TFT	Attention model using temporal context.
	TCN	Uses convolutions over time for sequences.
	TiDE	Fast MLP model for time series.
	N-BEATSx	Decomposes trend and includes external inputs.
	N-HiTS	Multi-scale hierarchy for accurate forecasts.

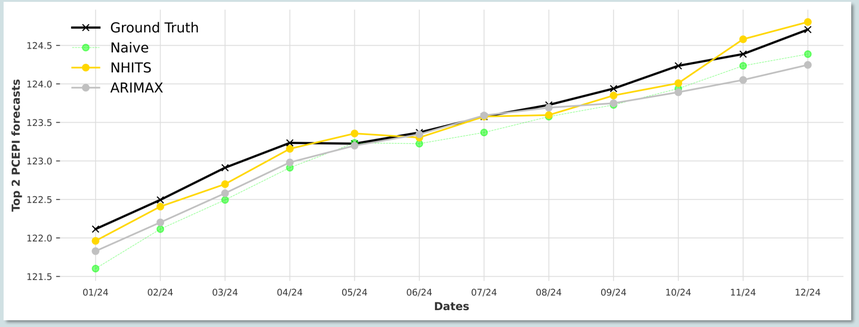
METHODOLOGY



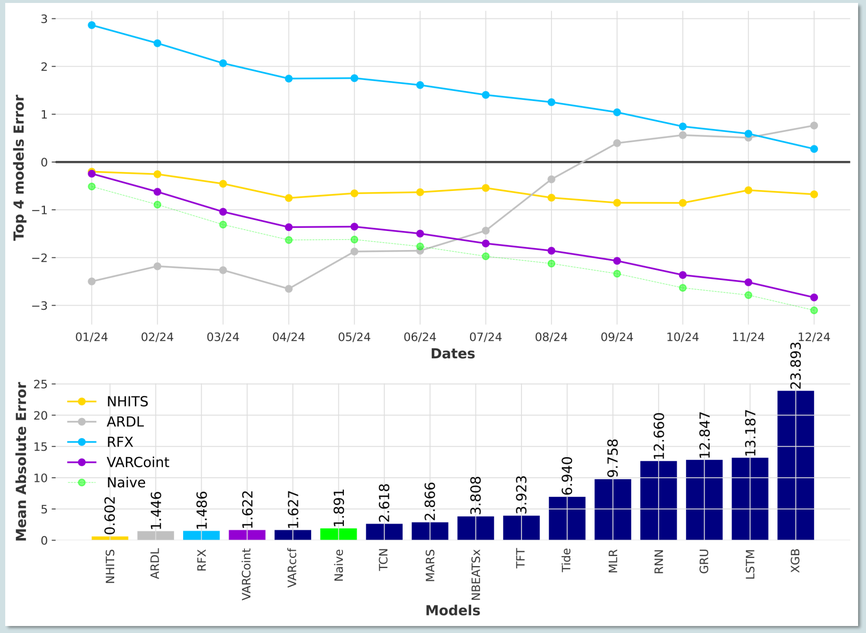
EVALUATION

HORIZON 1

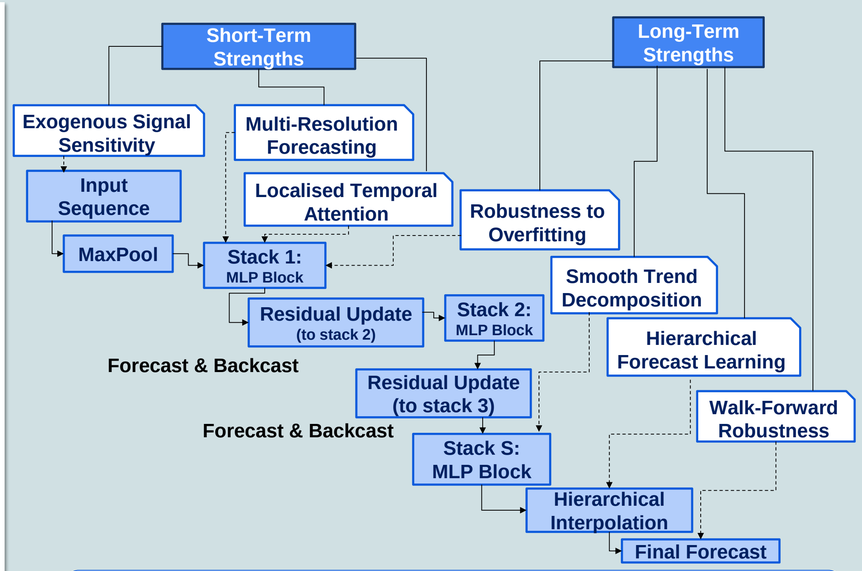




HORIZON 12



N-HiTS ARCHITECTURE



Key Finding: N-HiTS consistently outperformed statistical, machine learning, and other deep learning models across all horizons.

RESULTS

Short Term

- Inflation autocorrelation makes naive forecasting difficult to beat
- Complex models overfit short-term noise

Long Term

- Multi-step error accumulation causes model predictions to diverge
- Forecast horizon decay

Other Issues

- Global economy still recovering from COVID-19
- Governments and central banks set monetary policies to recovery
- Wars (China, Russia, US, Middle east)
- Elections or other political factors

Future Work

- Cross validation to find average error from different years.
- Use NLP/LLMs for news or tweets to extract relevant information.
- Political, socio-economic indicators
- Add trend-decomposed variables explicitly within models
- Ensemble models
- Adaptive models – auto trained using new data
- Hyperparameter optimising

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